LipSync
QUICKSTART GUIDE

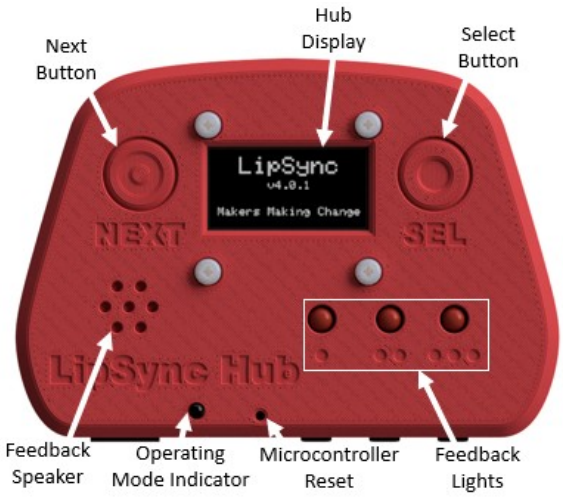
Makers Making Change
A Neil Squire Program

Introduction

The LipSync is a mouth-operated sip-and-puff joystick that allows a user with limited or no arm mobility to control a computer, gaming console, smart phone, or tablet. This Quickstart Guide is meant to briefly introduce the functions and use of the LipSync. For more detailed information, please consult the User Guide.

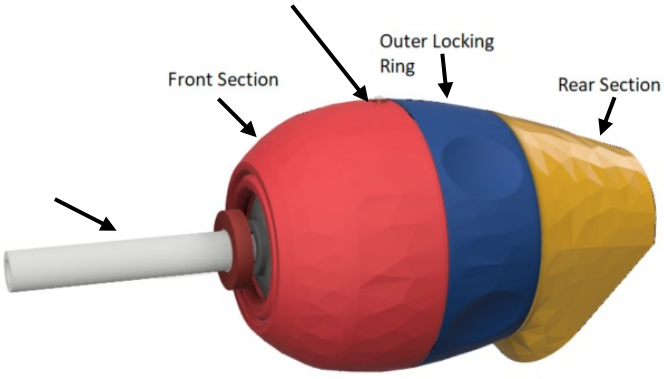
User Guide

Features











Setting up the LipSync

The following are the general steps needed to setup your LipSync device:

1. Mount LipSync Joystick
2. Mount LipSync Hub
3. Connect LipSync Interface Cable
4. Optional: Connect up to 3 Assistive Switches
5. Connect the LipSync Hub to the Host Device

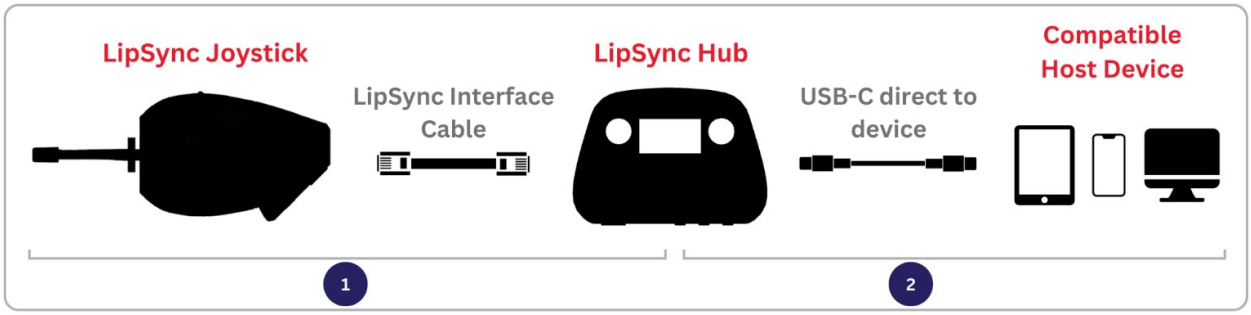
© 2024-2025 by Neil Squire.
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/LipSync

6. Once powered, wait until device initializes before moving joystick.

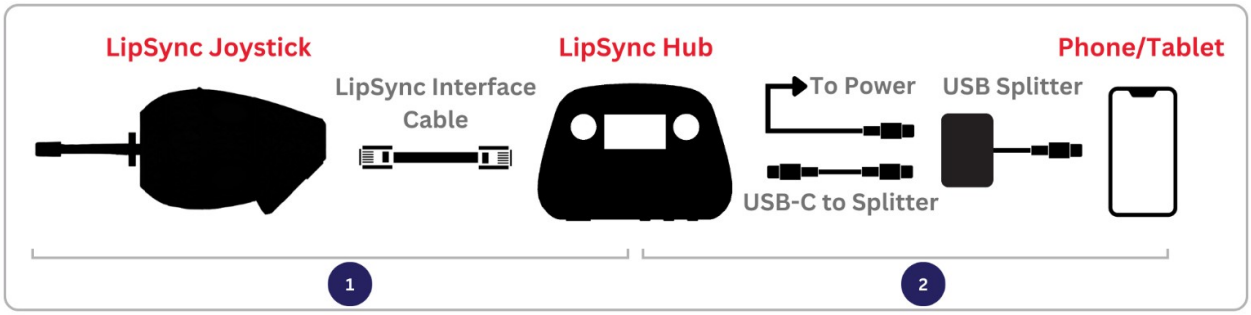
Connecting the LipSync to the Host Device

USB Connections

For direct USB connection to the host device, connect the appropriate USB cable from the LipSync Hub into the host device. Use an adapter if necessary.



For connection to the host device with an external power source, plug a power splitting adapter into the host device and connect the LipSync Hub and external power source to the splitter. Use an Adapter if necessary. **Required for iOS USB connection.**

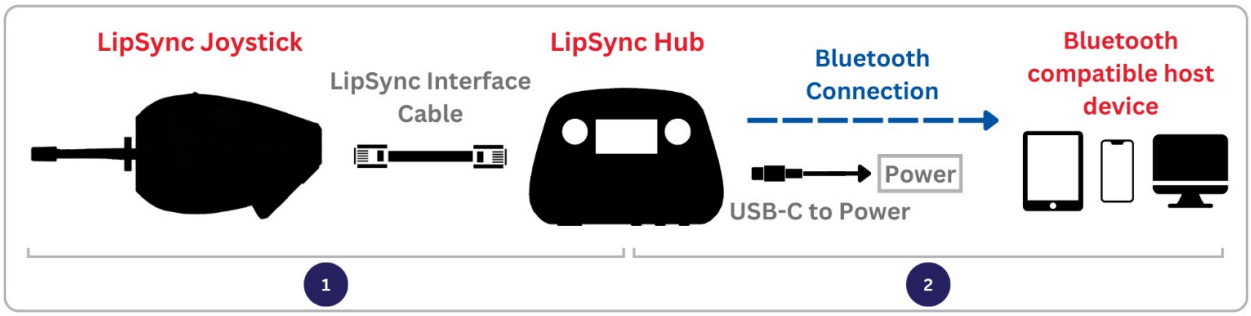


Wireless Connections

For wireless connection to the host device, plug the LipSync Hub directly into any power source using the appropriate adapter as required. Change the operating mode to BT Mouse through the Hub Menu
Find “LS_(DeviceID)” in the host device Bluetooth menu to pair and connect.

BY-SA © 2024-2025 by Neil Squire.
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/LipSync

LipSync
QUICKSTART GUIDE



Cursor Setup on iOS Devices

Open your device's ***Settings*** > ***Accessibility*** > ***Touch*** > ***AssistiveTouch*** > Turn it on.

Input Actions

Input			**Input Duration**	**Output** (based on Operating Mode)		
Sip and Puff	Assistive Switches	Hub Buttons		Mouse	Gamepad	Hub Menu
Puff	S1	Select	< 1 sec	Left click	Button 1	Select
			1-3 sec	Start Drag Mode	Button 3	-
			> 3 sec	Enter Hub Menu	Enter Hub Menu	Exit Hub Menu
-	S2	-	< 1 sec	Middle Click	Button 5	-
			1-3 sec	N/A	Button 6	-
			> 3 sec	Perform Center Reset	Perform Center Reset	-
Sip	S3	Next	< 1 sec	Right click	Button 2	Next
			1-3 sec	Start Scroll Mode	Button 4	-
			> 3 sec	Middle Click	Button 5	-
-	S1 + S3	Next + Select	< 1 sec	Enter Hub Menu	Enter Hub Menu	Exit Hub Menu
Joystick			-	Cursor movement	Joystick movement	

Hub Menu Settings

The Menu on the Hub Display provides a way to changing operating modes, adjusting settings, activate calibrations, and reset the LipSync device. See the Input Action table for how to access and navigate the menu. See the User Guide for more details.

Troubleshooting

Cursor Issues

If you are experiencing cursor drift, perform a center reset in one of the following ways:

Option A: Trigger Center Reset

Option B: Turn it off and then on again.

© 2024-2025 by Neil Squire.
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/LipSync

Either by:

- Activate the Hub Menu and navigate to **Center Reset**
- Using a very long press of assistive switch 2

Do not touch the joystick until “Ready to use” is displayed on the Hub Display.

- Disconnect/Reconnect the USB cable from the LipSync Hub.
- Restart the device through the Hub Menu..

Error Messages

If a hardware error is detected on startup, the LipSync will enter Safe Mode and display an error code. Refer to the User Guide for how to resolve the error. If issues persist, contact Support.

Support

For support, contact the maker of your device or Makers Making Change at info@makersmakingchange.com

© 2024-2025 by Neil Squire.
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/LipSync